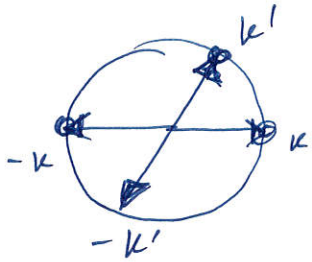
BCS

$$\hat{H}_{BCS} = \sum_{k\sigma} \xi_k c^+_{\vec{k}\sigma} c_{\vec{k}\sigma} + \sum_{\substack{kk' \\ \sigma\sigma'}} V_{kk'} c^+_{\vec{k}\sigma} c^+_{-\vec{k}-\sigma} c_{-\vec{k}'-\sigma'} c_{\vec{k}'\sigma'}$$



describes scattering of pairs of electrons with opposite momenta and spins over the Fermi surface.

Note that $\underbrace{(-\vec{k}, -\sigma)}_{\tilde{k}}$ is time-reversed of $\underbrace{(\vec{k}, \vec{\sigma})}_{k}$

denote:

$$\hat{H}_{BCS} = \sum_k \xi_k c^+_k c_k + \sum_{kk'} V_{kk'} c^+_k c^+_{\tilde{k}} c_{\tilde{k}'} c_{k'}$$

Physical reasoning: electrons form pairs $\Rightarrow$ expect anomalous (mean-field) averages to appear

$$F_k = \langle c_{\tilde{k}} c_k \rangle \quad , \quad F^+_k = \langle c^+_k c^+_{\tilde{k}} \rangle$$

Then approximate

$$\sum_{kk'} V_{kk'} \left[F^+_k + (c^+_k c^+_{\tilde{k}} - F^+_k) \right] \left[F_{k'} + (c_{\tilde{k}'} c_{k'} - F_{k'}) \right]$$

$$\approx \sum_{kk'} V_{kk'} \Big[F^+_k F_{k'} + (c^+_k c^+_{\tilde{k}} - F^+_k) F_{k'}$$

$$+ F^+_k (c_{\tilde{k}'} c_{k'} - F_{k'}) + \underbrace{\cancel{(c^+_k c^+_{\tilde{k}} - F^+_k)(c_{\tilde{k}'} c_{k'} - F_{k'})}}_{\text{neglect}} \Big]$$

Introduce $\Delta_k = -\sum_{k'} V_{kk'} F_{k'}$

Then
$$H_{BCS}^{MF} = \sum_k \xi_k c_k^\dagger c_k - \sum_k \left[\Delta_k^* c_{\bar{k}} c_k + \Delta_k c_k^\dagger c_{\bar{k}}^\dagger - \Delta_k F_k^\dagger \right]$$

annihilate and create pairs of particles
$\Rightarrow$ charge is not conserved!

This mean-field Hamiltonian is solved with the help of Bogoliubov transformation (unitary rotation)

$$\begin{pmatrix} \gamma_{k\uparrow} \\ \gamma_{-k\downarrow}^\dagger \end{pmatrix} = \begin{pmatrix} u_k^* & v_k \\ -v_k^* & u_k \end{pmatrix} \begin{pmatrix} c_{k\uparrow} \\ c_{-k\downarrow}^\dagger \end{pmatrix} = \begin{pmatrix} u_k^* c_{k\uparrow} + v_k c_{-k\downarrow}^\dagger \\ -v_k^* c_{k\uparrow} + u_k c_{-k\downarrow}^\dagger \end{pmatrix}$$

The functions u_k, v_k need to preserve fermion anticommutation relations

$$\Rightarrow \{\gamma_{k\uparrow}, \gamma_{p\uparrow}^\dagger\} = \delta_{kp}$$

$$\{ u_k^* c_{k\uparrow} + v_k c_{-k\downarrow}^\dagger , u_p c_{p\uparrow}^\dagger + v_p^* c_{-p\downarrow} \} =$$

$$= u_k^* u_p \underbrace{\{ c_{k\uparrow}, c_{p\uparrow}^\dagger \}}_{\delta_{kp}} + v_k v_p^* \underbrace{\{ c_{-k\downarrow}^\dagger , c_{-p\downarrow} \}}_{\delta_{kp}} =$$

$$= \left(|u_k|^2 + |v_k|^2 \right) \delta_{kp} = \delta_{kp}$$

$$\Rightarrow |u_k|^2 + |v_k|^2 = 1 \Rightarrow \det \begin{pmatrix} u_k^* & v_k \\ -v_k^* & u_k \end{pmatrix} = 1 .$$

Inverse $\begin{pmatrix} c_{k\uparrow} \\ c^{+}_{-k\downarrow} \end{pmatrix} = \begin{pmatrix} u_k & -v_k \\ v_k^{*} & u_k^{*} \end{pmatrix} \begin{pmatrix} \gamma_{k\uparrow} \\ \gamma^{+}_{-k\downarrow} \end{pmatrix}$

Observe that

$$H^{MF}_{BCS} = \sum_k \left(c^{+}_{k\uparrow}, c_{-k\downarrow} \right) \begin{pmatrix} \xi_k & -\Delta_k \\ -\Delta_k^{*} & -\xi_{-k} \end{pmatrix} \begin{pmatrix} c_{k\uparrow} \\ c^{+}_{-k\downarrow} \end{pmatrix}$$

Here $\left(c^{+}_{k\uparrow}, c_{-k\downarrow} \right) = \begin{pmatrix} c_{k\uparrow} \\ c^{*}_{-k\downarrow} \end{pmatrix}^{+} = \left\{ \begin{pmatrix} u_k & -v_k \\ v_k^{*} & u_k^{*} \end{pmatrix} \begin{pmatrix} \gamma_{k\uparrow} \\ \gamma^{+}_{-k\downarrow} \end{pmatrix} \right\}^{+}$

$$= \begin{pmatrix} \gamma_{k\uparrow} \\ \gamma^{+}_{-k\downarrow} \end{pmatrix}^{+} \underbrace{\begin{pmatrix} u_k & -v_k \\ v_k^{*} & u_k^{*} \end{pmatrix}^{+}}_{\begin{pmatrix} u_k & v_k^{*} \\ -v_k & u_k^{*} \end{pmatrix}^{*} = \begin{pmatrix} u_k^{*} & v_k \\ -v_k^{*} & u_k \end{pmatrix}} = \left(\gamma^{+}_{k\uparrow}, \gamma_{-k\downarrow} \right) \begin{pmatrix} u_k^{*} & v_k \\ -v_k^{*} & u_k \end{pmatrix}$$

$$\Rightarrow H^{MF}_{BCS} = \sum_k \left(\gamma^{+}_{k\uparrow}, \gamma_{-k\downarrow} \right) \begin{pmatrix} u_k^{*} & v_k \\ -v_k^{*} & u_k \end{pmatrix} \begin{pmatrix} \xi_k & -\Delta_k \\ -\Delta_k^{*} & -\xi_k \end{pmatrix} \cancel{\begin{pmatrix} u_k^{*} & v_k \\ -v_k^{*} & u_k \end{pmatrix}}$$

$$\cdot \begin{pmatrix} u_k & -v_k \\ v_k^{*} & u_k^{*} \end{pmatrix} \begin{pmatrix} \gamma_{k\uparrow} \\ \gamma^{+}_{-k\downarrow} \end{pmatrix}$$

$$\begin{pmatrix} u_k^* & v_k \\ -v_k^* & u_k \end{pmatrix} \begin{pmatrix} \xi_k & -\Delta_k \\ -\Delta_k^* & -\xi_k \end{pmatrix} \begin{pmatrix} u_k & -v_k \\ v_k^* & u_k^* \end{pmatrix} =$$

$$= \begin{pmatrix} u_k^* & v_k \\ -v_k^* & u_k \end{pmatrix} \begin{pmatrix} \xi_k u_k - \Delta_k v_k^* & -\xi_k v_k - \Delta_k u_k^* \\ -u_k \Delta_k^* - \xi_k v_k^* & \Delta_k^* v_k - \xi_k u_k^* \end{pmatrix} =$$

$$= \begin{pmatrix} \xi_k \left(|u_k|^2 - |v_k|^2\right) - \Delta_k u_k^* v_k^* - \Delta_k^* u_k v_k ; & \left(-\Delta_k \bar{u}_k^2 + \Delta_k^* v_k^2 - 2\xi_k \bar{u}_k v_k\right) \\ \left(\Delta_k \bar{v}_k^2 - \Delta_k^* u_k^2 - 2\xi_k \bar{v}_k u_k\right); & \xi_k \left(|v_k|^2 - |u_k|^2\right) + \Delta_k \bar{v}_k \bar{u}_k + \Delta_k^* u_k v_k \end{pmatrix}$$

<u>Require</u> off-diagonal elements to be <u>zero</u>

(consider here for simplicity <u>real</u> functions)

$$-\Delta_k \left(u_k^2 - v_k^2\right) - 2\xi_k u_k v_k = 0$$

$$\Delta^2 \underbrace{\left(u_k^2 - v_k^2\right)^2}_{\left(1 - 2v_k^2\right)^2} = 4\xi_k^2 u_k^2 v_k^2 = 4\xi_k^2 \left(1 - v_k^2\right) v_k^2$$

Use $u_k^2 + v_k^2 = 1$

$$\Rightarrow \left(1 - 2v_k^2\right)^2 = \frac{4\xi_k^2}{\Delta^2} \left(1 - v_k^2\right) v_k^2$$

$$1 - 4v_k^2 + 4v_k^4 = \frac{4\xi^2}{\Delta^2} \left(v_k^2 - v_k^4\right)$$

$$4v_k^4\left(1+\frac{\xi^2}{\Delta^2}\right)-4v_k^2\left(1+\frac{\xi^2}{\Delta^2}\right)+1=0$$

$$4v_k^4-4v_k^2+\frac{\Delta^2}{\Delta^2+\xi^2}=0$$

$$v_k^2=\left(2\pm\sqrt{4-\frac{4\Delta^2}{\Delta^2+\xi^2}}\right)\frac{1}{4}=\frac{1}{2}\left(1\pm\frac{\xi_k}{\sqrt{\xi_k^2+\Delta^2}}\right)$$

$$\Rightarrow u_k^2=\frac{1}{2}\left(1\mp\frac{\xi_k}{\sqrt{\xi_k^2+\Delta_k^2}}\right)$$

$$u_k^2v_k^2=\frac{1}{4}\left(1-\frac{\xi_k^2}{\xi_k^2+\Delta^2}\right)=\frac{\Delta^2}{4(\xi_k^2+\Delta^2)}\Rightarrow u_kv_k=\frac{\pm\Delta_k}{2\sqrt{\xi_k^2+\Delta_k^2}}$$

First diagonal element

$$\xi_k(u_k^2-v_k^2)-2\Delta_k u_kv_k=\xi_k\left(\frac{\mp\xi_k}{\sqrt{\xi_k^2+\Delta_k^2}}\right)-2\Delta\frac{\pm\Delta_k}{2\sqrt{\xi_k^2+\Delta_k^2}}$$

$$=\mp\frac{(\xi_k^2+\Delta_k^2)}{\sqrt{\xi_k^2+\Delta_k^2}}\underset{\text{must be}}{\Longrightarrow}+E_k=\sqrt{\xi_k^2+\Delta_k^2}$$

$\Downarrow$

choose lower sign:

$$\begin{cases} v_k^2=\frac{1}{2}\left(1-\frac{\xi_k}{E_k}\right)\\ u_k^2=\frac{1}{2}\left(1+\frac{\xi_k}{E_k}\right)\end{cases}$$

So with these algebra sorted out we have

$$H_{BCS}^{MF} = \sum_k \left(\gamma^\dagger_{k\uparrow}, \gamma_{-k\downarrow}\right) \begin{pmatrix} E_k & 0 \\ 0 & -E_k \end{pmatrix} \begin{pmatrix} \gamma_{k\uparrow} \\ \gamma^\dagger_{-k\downarrow} \end{pmatrix}$$

$$= \sum_k E_k \left(\gamma^\dagger_{k\uparrow}\gamma_{k\uparrow} + \gamma^\dagger_{-k\downarrow}\gamma_{-k\downarrow}\right)$$

$$= \sum_k E_k \left(\gamma^\dagger_{k\uparrow}\gamma_{k\uparrow} + \gamma^\dagger_{k\downarrow}\gamma_{k\downarrow}\right)$$

with $E_k = \sqrt{\xi_k^2 + \Delta_k^2}$, $\xi_k = \frac{\hbar^2 k^2}{2m} - \varepsilon_F$

What is Δ?

$$\Delta_k = -\sum_{k'} V_{kk'} F_{k'}$$

$$\hookrightarrow \langle c_{-k'\downarrow} c_{k'\uparrow}\rangle$$

$$c_{k\uparrow} = u_k \gamma_{k\uparrow} - v_k \gamma^\dagger_{-k\downarrow}$$

$$c_{-k\downarrow} = \left(c^\dagger_{-k\downarrow}\right)^\dagger = v_k \gamma^\dagger_{k\uparrow} + u_k \gamma_{-k\downarrow}$$

$$\Rightarrow \langle c_{-k\downarrow} c_{k\uparrow}\rangle = \langle (v_k \gamma^\dagger_{k\uparrow} + u_k \gamma_{-k\downarrow})(u_k \gamma_{k\uparrow} - v_k \gamma^\dagger_{-k\downarrow})\rangle$$

$$= u_k v_k \left(\langle \gamma^\dagger_{k\uparrow}\gamma_{k\uparrow}\rangle - \underbrace{\langle \gamma_{-k\downarrow}\gamma^\dagger_{-k\downarrow}\rangle}_{1 - \langle \gamma^\dagger_{-k\downarrow}\gamma_{-k\downarrow}\rangle}\right) =$$

$$= -\left(1 - \langle \gamma^\dagger_{k\uparrow}\gamma_{k\uparrow}\rangle - \langle \gamma^\dagger_{-k\downarrow}\gamma_{-k\downarrow}\rangle\right) u_{k'} v_{k'}$$

so that $\Delta_k = \sum_{k'} V_{kk'} u_{k'} v_{k'} \left(1 - 2\langle \gamma^\dagger_{k'\uparrow}\gamma_{k'\uparrow}\rangle\right).$

But $u_k v_k = \dfrac{+\Delta}{2\sqrt{\xi_k^2+\Delta^2}} = \dfrac{\ominus\Delta}{2E_k}$

$$\langle \gamma^+_{k\uparrow}\gamma_{k\uparrow}\rangle = \frac{1}{e^{E_k/k_BT}+1}$$ = Fermi-Dirac distribution function of γ-particles

$V_{kk'} = -V < 0$ for k, k' within shell $\hbar\omega_D$ of Fermi surface.

$$\Delta = -V\sum_k \frac{\ominus\Delta}{2E_k}\left(1 - \frac{2}{e^{\frac{E_k}{T}}+1}\right)$$

At $T=0$:

$$\Delta = V\sum_k \frac{\Delta}{2E_k} \Rightarrow 1 = \frac{V}{2}\int\frac{d\vec{k}}{(2\pi)^3}\frac{1}{\sqrt{\xi_k^2+\Delta^2}} =$$

$$= \frac{V}{2}N_0\int_{-\hbar\omega_D}^{\hbar\omega_D}\frac{d\xi}{\sqrt{\xi^2+\Delta^2}}$$

$$= VN_0\int_0^{\hbar\omega_D}\frac{d\xi}{\sqrt{\xi^2+\Delta^2}}$$

$$\int_0^{\hbar\omega_D}\frac{d\xi}{\sqrt{\xi^2+\Delta^2}} = \ln\left(\xi+\sqrt{\xi^2+\Delta^2}\right)\Big|_0^{\hbar\omega_D} \cong \ln\frac{2\hbar\omega_D}{\Delta}$$

$$\frac{1}{VN_0} = \ln\frac{2\hbar\omega_D}{\Delta} \Rightarrow \Delta = 2\hbar\omega_D\, e^{-1/N_0V} \quad !$$

$$1 = \frac{VN_0}{2}\int_{-\hbar\omega_D}^{\hbar\omega_D} \frac{d\xi}{E_k} \frac{e^{E_k/T}-1}{e^{E_k/T}+1} = N_0 V \int_0^{\hbar\omega_D} d\xi \frac{\tanh\left(\frac{E_k}{2T}\right)}{E_k}$$

Re-write as

$$1 + V\sum_k \frac{\Delta}{2E_k} = 2V\int d\xi \frac{1}{E_k\left(e^{E_k/T}+1\right)}$$

$$\frac{2\Delta(0)}{k_B T_c} \approx 3.53$$